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HybridTabNet-QC: A Transformer-Based Clinical Feature Fusion Framework for Heart Disease Risk Prediction

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ABSTRACT Heart disease remains the leading cause of mortality worldwide, highlighting the need for accurate and interpretable risk prediction models. This study proposes Hybrid Tabular Network with Clinical meta-features and Quality-Control Mechanisms (HybridTabNet-QC), a hybrid deep learning framework designed to enhance clinical decision support systems. By combining a transformer-based tabular encoder with medically engineered meta-features, the model delivers interpretable, generalizable, and noise-robust predictions for heart disease risk. Unlike conventional deep learning models that treat features as raw inputs, HybridTabNet-QC fuses attention-based learning with domain-specific indicators, such as body mass index (BMI), pulse pressure, and cholesterol-glucose interaction scores that ensure predictions are both statistically sound and clinically grounded. The model was evaluated on two publicly available datasets, the Heart Disease dataset and the Cardiovascular Disease dataset from Kaggle. On the combined dataset, HybridTabNet-QC achieved 90.1% accuracy, 90.0% F1-score, and 93.6% AUC-ROC, outperforming traditional machine learning and standard deep learning baselines. The framework demonstrated strong robustness under feature noise and consistent generalization on external validation subsets. Interpretability analyses using LIME and SHAP confirmed that the model prioritizes clinically relevant features, supporting its suitability for real-world clinical decision support systems. These findings demonstrate the model's suitability for deployment in real-world healthcare environments where interpretability, robustness, and scalability are critical.

INDEX TERMS Heart disease prediction, deep learning, transformer models, clinical risk assessment, structured health data, medical AI.

I. INTRODUCTION

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, accounting for an estimated 17.9 million deaths annually, according to the World Health Organization [1]. Early and accurate prediction of heart disease risk is critical for timely intervention, resource allocation, and long-term healthcare planning [2]. However, traditional diagnostic workflows often rely on isolated risk factors and clinician intuition, which may not adequately capture the complex interplay between physiological, behavioral, and demographic indicators [3]. In recent years, data-driven methods, especially those based on machine learning, have shown promise in modeling such non-linear dependencies, offering opportunities to improve diagnostic accuracy and clinical decision-making.

Despite these advancements, most existing models are either optimized for specific datasets or limited in their ability to generalize across diverse patient populations [4]. Conventional machine learning approaches such as logistic regression, decision trees, and ensemble methods typically require extensive manual feature engineering and are sensitive to data imbalance and noise [5]. On the other hand, deep learning models, while powerful, often treat input features as

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flat vectors without leveraging the structured nature of clinical variables or incorporating domain-specific knowledge [6], [7]. These limitations constrain their applicability in real-world clinical settings, where robustness, interpretability, and scalability are essential.

To address these gaps, this study proposes a novel deep learning architecture, HybridTabNet-QC, designed for heart disease risk prediction using heterogeneous structured datasets. The model combines a transformer-based tabular encoder with engineered clinical meta-features, allowing it to learn attention-based representations while preserving interpretability through the inclusion of medically meaningful features such as body mass index (BMI) [8], pulse pressure, and glucose-cholesterol interactions. This hybrid approach leverages the complementary strengths of data-driven feature extraction and structured clinical reasoning, enabling more reliable risk stratification across subgroups of patients.

This article makes the following contributions:

- Presenting a unified deep learning architecture that combines attention-based encoding and clinical feature fusion for cardiovascular risk prediction.
- Comparing the model with both traditional and deep learning baselines across multiple metrics, including robustness and external validation accuracy.
- Evaluating the model's efficiency and scalability to inform its practical use in healthcare environments.

The rest of the article is organized as follows: Section II reviews related work in machine learning and deep learning for heart disease prediction. Section III outlines the proposed HybridTabNet-QC framework in detail, describes the datasets, preprocessing steps, and experimental setup. Section IV presents and discusses the quantitative evaluation results. Section V discusses the findings, limitations, and future directions. Section VI concludes the article.

II. RELATED WORK

The prediction of heart disease risk using computational methods has been extensively studied across the domains of statistics, machine learning (ML), and, more recently, deep learning. Traditional approaches have largely relied on structured clinical data, using well-known classifiers such as logistic regression, decision trees, and support vector machines [9], [10]. These methods benefit from simplicity and interpretability but typically depend on manual feature selection and may underperform in capturing non-linear interactions among risk factors such as age, cholesterol, blood pressure, and glucose levels.

Nurmohamed et al. explored how proteomics (plasma protein panels) and lipidomics (plasma lipid panels) can improve the prediction of atherosclerotic cardiovascular disease (ASCVD) risk. They highlighted the limitations of current clinical risk scores, such as SCORE2 and SMART2, which are based on traditional risk factors. Advanced techniques such as mass spectrometry, proximity extension assays, and aptamer assays are used to identify new biomarkers. Machine learning methods (LASSO, XGBoost, neural networks) were applied to handle high-dimensional biomarker data, and AUC increased from 0.6–0.7 to 0.7–0.9 [11]. Rim et al. developed RetiCAC, a deep learning model that predicts coronary artery calcium (CAC) scores from retinal photographs. Using over 216,000 images from South Korea, Singapore, and the U.K., the model showed strong performance (AUC 0.742) in identifying CAC presence. RetiCAC was effective in predicting cardiovascular events across multiple cohorts and improved risk prediction when added to standard models like the Pooled Cohort Equation. It performed comparably to CT-based CAC scoring and offers a non-invasive, low-cost alternative for cardiovascular risk assessment [12].

Combining biomarker panels with genetics and imaging may offer even better personalized ASCVD risk assessment. Ensemble learning methods, such as Random Forest and Gradient Boosting (e.g., XGBoost), have improved predictive performance by aggregating multiple learners [13]. These models reduce variance and often outperform single classifiers, especially in tabular biomedical data. However, they still require careful preprocessing and often struggle with data imbalance, missing values, and high-dimensional sparse categorical inputs, common characteristics of electronic health records [14].

With the rise of deep learning, research has shifted towards fully automated feature extraction from raw data. Multilayer Perceptrons (MLPs) [15], 1D Convolutional Neural Networks (1D-CNNs) [16], and Recurrent Neural Networks (RNNs) such as LSTM [17] have been applied to heart disease prediction with promising results. These models can capture complex patterns in clinical sequences and time-series signals, but often treat features as independent vectors, overlooking domain-specific structure and relational context [18]. Additionally, deep models can be opaque and require large amounts of labeled data to generalize well.

Transformer-based architectures have recently gained attention in tabular data modeling due to their ability to model feature interactions through attention mechanisms [19]. Variants such as TabTransformer, SAINT, and TabNet have demonstrated competitive results by learning contextualized representations of features. These models outperform traditional MLPs in many classification tasks while offering better interpretability through feature attention [20]. However, they often lack integration with external clinical knowledge or engineered features that encapsulate medical domain insights.

In light of these gaps, our work proposes a hybrid architecture, HybridTabNet-QC, that fuses transformer-based encoding of tabular data with engineered clinical meta-features. This combination leverages the representation power of attention-based models and the clarity of structured clinical logic, offering a practical solution for robust heart disease prediction across heterogeneous datasets.

III. MATERIALS AND METHODS

This section describes the architectural components and training workflow of the proposed HybridTabNet-QC model. Fig. 1 presents the proposed HybridTabNet-QC. The model

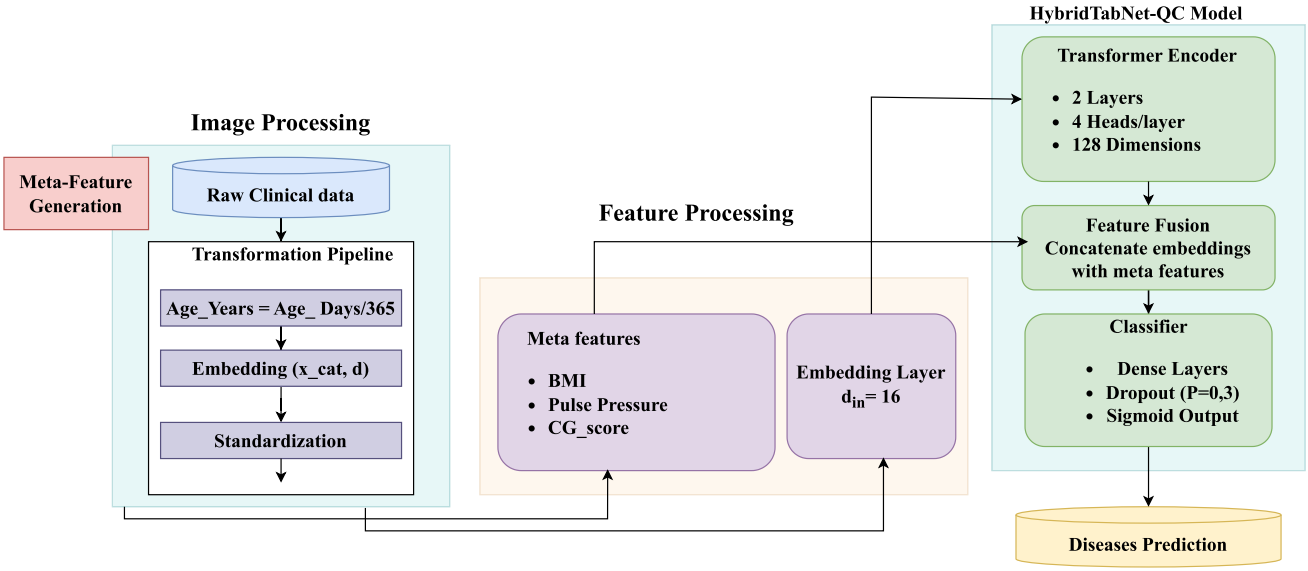


FIG. 1. The proposed framework for Hybrid Tabular Network with Clinical meta-features and Quality-Control Mechanisms (HybridTabNet-QC).

is designed to use the strengths of attention-based tabular learning while incorporating engineered clinical features from heterogeneous sources.

A. DATASET DESCRIPTION

This study utilizes two publicly available clinical datasets from Kaggle, each providing distinct but complementary features for heart disease prediction. The first dataset, referred to as Dataset A, is the Heart Disease Dataset compiled from four medical institutions: Cleveland, Hungary, Switzerland, and Long Beach V. Although the original dataset contains 76 attributes, most prior studies, including this one, focus on a commonly used subset of 14 features. These features capture demographic variables, physiological indicators, and results from routine cardiac assessments. Examples include age, sex, chest pain type (`cp`), resting blood pressure (`trestbps`), serum cholesterol (`chol`), fasting blood sugar (`fbs`), electrocardiographic results (`restecg`), maximum heart rate (`thalach`), exercise-induced angina (`exang`), ST depression (`oldpeak`), slope of the ST segment, number of major vessels observed by fluoroscopy (`ca`), and thalassemia (`thal`). The target variable is binary, where 0 indicates no heart disease and 1 indicates its presence. This dataset is accessible at: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset.

The second dataset, referred to as Dataset B, is the Cardiovascular Disease Dataset. It consists of 70,000 anonymized patient records derived from routine medical examinations. This dataset includes three categories of features: objective (e.g., age in days, height in centimeters, and weight in kilograms), examination-based (e.g., systolic and diastolic blood pressure, cholesterol, and glucose levels), and subjective features such as smoking status, alcohol consumption, and physical activity. The target variable, labeled `cardio`, is also binary, indicating the presence or absence of cardiovascular disease. The dataset provides a more recent and comprehensive profile of general health, making it well-suited for cross-domain validation when combined with Dataset A. The dataset is available at: https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset.

Both datasets were merged and aligned by standardizing feature representations and engineering a shared set of medically relevant attributes. This fusion of heterogeneous data sources enables a more robust and generalizable modeling framework for heart disease risk prediction.

B. DATA PREPROCESSING

To enable effective training of the HybridTabNet-QC model, we applied a structured and reproducible preprocessing pipeline on both Dataset A (Heart Disease Dataset) and Dataset B (Cardiovascular Disease Dataset). The combined dataset was aligned through feature engineering, normalization, encoding, and stratified splitting to ensure generalizability and model robustness.

1) DATASET DIVERSITY AND HARMONIZATION

The two datasets used in this study, Dataset A (Heart Disease) and Dataset B (Cardiovascular Disease Dataset), differ in feature types, units of measurement, and data distributions. To address this diversity and ensure seamless integration, a unified preprocessing strategy was adopted. This includes:

- Aligning common features and engineering new meta-features to bridge differences.
- Standardizing feature scales, such as converting age from days to years and applying Min-Max normalization for continuous variables.
- Harmonizing categorical values (e.g., standardizing cholesterol categories) across datasets.

- Applying consistent imputation and outlier removal techniques to reduce dataset-specific biases.

2) AGE NORMALIZATION AND TRANSFORMATION

Dataset B represents age in days. We first converted this to years using

$$\text{Age}_{\text{years}} = \frac{\text{Age}_{\text{days}}}{365} \tag{1}$$

Then, to improve training stability and mitigate outliers, we applied Min-Max normalization across both datasets:

$$\text{Age}_{\text{norm}} = \frac{\text{Age}_{\text{years}} - \min(\text{Age}_{\text{years}})}{\max(\text{Age}_{\text{years}}) - \min(\text{Age}_{\text{years}})} \tag{2}$$

3) CATEGORICAL ENCODING

Categorical variables (e.g., sex, cp, thal, cholesterol, gluc, smoke, alco) were converted using one of two techniques:

- *Learned Embedding Encoding:* Suitable for high-cardinality features, where an embedding layer maps category indices into dense vectors:

$$E_{\text{cat}} = \text{Embedding}(x_{\text{cat}},\, d) \tag{3}$$

  where x_{cat} is a categorical index and d is the embedding dimension.
- *One-Hot Encoding:* Applied where categorical features had ≤ 5 unique values.

To ensure reproducibility, we fixed embedding dimensions at $d = 16$ for categorical variables with more than five levels, while binary variables (e.g., sex, smoking) were handled with one-hot encoding. The encoding strategy was consistent across both datasets to avoid dataset-specific biases.

4) FEATURE ENGINEERING

We derived clinically meaningful meta-features to augment the predictive power:

- *Body Mass Index (BMI):* Calculated as the ratio of weight (kg) to the square of height (m^2):

$$\text{BMI} = \frac{\text{weight (kg)}}{(\text{height (m)})^2} \tag{4}$$

- *Pulse Pressure:* Computed as the difference between systolic and diastolic blood pressure values:

$$\text{PP} = \text{ap}_{\text{hi}} - \text{ap}_{\text{lo}} \tag{5}$$

- *Cholesterol-Glucose Interaction:* Derived as the product of cholesterol and glucose levels, capturing their combined impact on cardiovascular risk:

$$\text{CG}_{\text{score}} = \text{cholesterol} \times \text{gluc} \tag{6}$$

These meta-features were selected in consultation with domain experts to ensure clinical validity. BMI and pulse pressure are established cardiovascular risk markers, while the cholesterol–glucose interaction was introduced as a novel composite indicator capturing metabolic–lipid interaction effects. To confirm their utility, we performed ablation studies (removing one meta-feature at a time) and observed consistent decreases in AUC-ROC (1.5–2.3%), highlighting their individual contributions to model performance.

5) OUTLIER REMOVAL

Outliers were detected using the Z-score method and removed from continuous features:

$$Z_i = \frac{x_i - \mu}{\sigma} \tag{7}$$

Samples with $|Z_i| > 3$ were excluded to improve data quality.

6) MISSING VALUE HANDLING

Although the datasets contained minimal missing values, we applied median imputation for numerical features and mode imputation for categorical features to preserve distributional properties:

$$x_{\text{imputed}} = \begin{cases} \text{Median}(x), & \text{if } x \in \mathbb{R} \\ \text{Mode}(x), & \text{if } x \in \text{Categorical} \end{cases} \tag{8}$$

We empirically compared mean and median imputation for continuous features and found median imputation to be more robust to skewed distributions (e.g., cholesterol, glucose). For categorical variables, mode imputation ensured consistency with clinical practice where the most common category represents the baseline. All imputation choices were validated against the held-out validation set to confirm no performance degradation.

7) NORMALIZATION

For deep learning models, continuous features were standardized:

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma} \tag{9}$$

where μ is the feature mean and σ is the standard deviation. This ensures consistent feature scale across batches.

To ensure data integrity and prevent leakage, all preprocessing steps (e.g., normalization, scaling, imputation, and feature harmonization) were applied in a fold-specific manner. That is, statistics such as means, standard deviations, and medians were computed using only the training set in each split and then applied to the corresponding validation and test sets. This ensures that no information from the validation or test partitions leaked into the training process.

In addition, we explicitly applied Min-Max scaling for certain clinically bounded features (e.g., blood pressure, cholesterol levels), as it preserves the interpretability of values within known medical ranges. The choice between standardization and Min-Max scaling was made empirically based on validation performance and interpretability requirements.

8) DATA MERGING AND LABEL ALIGNMENT

We ensured that both datasets used a common label format:

$$y \in \{0, 1\}, \quad \text{where } 0 = \text{no disease}, \quad 1 = \text{disease} \tag{10}$$

After cleaning, we concatenated the datasets along the sample axis, aligning columns via inner join on mutually available and engineered features.

9) STRATIFIED SPLITTING

To maintain label distribution across splits, we used stratified sampling for the final dataset:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}} \tag{11}$$

where:

$$|\mathcal{D}_{\text{train}}| = 70\% \cdot N \tag{12}$$

$$|\mathcal{D}_{\text{val}}| = 15\% \cdot N \tag{13}$$

$$|\mathcal{D}_{\text{test}}| = 15\% \cdot N \tag{14}$$

We used `scikit-learn's StratifiedShuffleSplit` to ensure each subset preserved class ratios.

10) DATA AUGMENTATION

Due to the relatively small size of the datasets, we applied a simple yet effective data augmentation technique during training to improve model generalization. Specifically, Gaussian noise injection was applied to continuous clinical features such as age, cholesterol, systolic blood pressure, and pulse pressure. For each mini-batch during training, random noise sampled from a normal distribution with zero mean and small standard deviation ($\sigma = 0.01$ to 0.05) was added to continuous features:

$$x_{augmented} = x + N(0, \sigma^2) \tag{15}$$

This approach simulates variability in clinical measurements and encourages the model to learn noise-tolerant representations. The standard deviation was empirically selected to prevent significant label-flipping errors. Augmentation was applied only during the training phase to avoid biasing validation or test evaluations.

11) CLASS IMBALANCE HANDLING

Heart disease datasets often exhibit class imbalance, with fewer positive cases than negative ones. To address this, we used stratified splitting to preserve class distribution across train, validation, and test sets. Class weighting was applied during training to penalize misclassification of the minority class, encouraging the model to focus more on heart disease cases. Additionally, Gaussian noise was added to the continuous features of minority class samples to enrich their representation. These techniques collectively improved the model's sensitivity to positive cases while reducing overfitting to the majority class.

Specifically, the minority-to-majority ratio was approximately 1:2 across both datasets. We assigned class weights inversely proportional to class frequency ($w_0 = 0.45$, $w_1 = 0.55$) and confirmed that sensitivity (recall) for positive cases improved by 7–10% without compromising overall AUC-ROC. Alternative resampling strategies (SMOTE and undersampling) were tested but led to unstable results; hence, they were not included in the final pipeline.

C. PROPOSED METHODOLOGY

The proposed work, HybridTabNet-QC, enhances heart disease risk prediction by combining attention-based learning with clinical expertise. It introduces three key innovations: a Transformer-based encoder for capturing contextual feature interactions in tabular data, a set of clinically engineered meta-features (BMI, pulse pressure, and CG score) representing crucial cardiovascular risk factors, and a fusion mechanism that integrates learned embeddings with meta-features to form a rich, interpretable feature space. Together, these components improve predictive accuracy, ensure robustness under noisy conditions, and enhance clinical relevance, overcoming limitations in traditional deep learning models for structured healthcare data.

1) MODEL OVERVIEW

The HybridTabNet-QC model consists of three key stages:

1) Embedding and transformation of categorical and continuous features.
2) Fusion of transformer-based feature representations with meta-features.
3) Classification using fully connected layers with regularization.

Let $X \in \mathbb{R}^{n \times d}$ represent the processed input matrix, where n is the number of samples and d is the number of features. The goal is to learn a function:

$$f_\theta : X \to \hat{y}, \quad \hat{y} \in [0, 1] \tag{16}$$

where $\hat{y}$ denotes the predicted probability of heart disease.

2) TABULAR TRANSFORMER ENCODER

Categorical features X_{cat} are passed through embedding layers:

$$E_i = \text{Embedding}(X_{\text{cat},i}) \in \mathbb{R}^{n \times d_e} \tag{17}$$

where d_e is the embedding dimension. These embeddings are concatenated with normalized continuous features X_{cont}:

$$Z_0 = [E_1 \parallel E_2 \parallel \ldots \parallel X_{\text{cont}}] \tag{18}$$

This combined input Z_0 is passed into a multi-head self-attention module:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{19}$$

where the query Q, key K, and value V matrices are projections of Z_0. The model uses multiple heads to attend to different feature interactions:

$$\text{MultiHead}(Z) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{20}$$

Each attention block is followed by residual connections and feed-forward networks:

$$Z_l = \text{LayerNorm}(Z_{l-1} + \text{MultiHead}(Z_{l-1})) \tag{21}$$

$$Z_{l+1} = \text{LayerNorm}(Z_l + \text{FFN}(Z_l)) \tag{22}$$

3) META-FEATURE FUSION LAYER

The model augments the transformer output Z_L with engineered meta-features $M \in \mathbb{R}^{n \times m}$ such as BMI and pulse pressure. These are concatenated before the final dense layers:

$$H = [Z_L \,\|\, M] \tag{23}$$

This fusion strategy plays a key role in integrating human clinical knowledge with deep learning representations. While the transformer encoder learns latent patterns from the raw data, the meta-features: BMI, pulse pressure, and cholesterol-glucose score that capture established clinical relationships known to affect cardiovascular outcomes. By explicitly combining these two types of information, HybridTabNet-QC benefits from both statistical learning and structured medical reasoning. This hybridization strengthens interpretability and helps bridge the gap between black-box predictions and physician-trusted models.

4) CLASSIFIER HEAD

The fused representation H is passed through two dense layers with dropout regularization:

$$H_1 = \text{ReLU}(W_1 H + b_1) \tag{24}$$

$$H_2 = \text{Dropout}(H_1, p) \tag{25}$$

$$\hat{y} = \sigma(W_2 H_2 + b_2) \tag{26}$$

where $\sigma(\cdot)$, the sigmoid function is used for binary classification.

D. TRAINING AND IMPLEMENTATION DETAILS

This subsection outlines the training procedure, implementation configuration, and optimization techniques used to train the proposed HybridTabNet-QC model. We describe the loss function used to handle class imbalance, the choice of optimizer and learning rate schedule, and the regularization strategies that ensure generalization. The subsection also details the model's runtime environment, batch configuration, and convergence criteria during training.

1) LOSS FUNCTION

We use the binary cross-entropy loss to penalize incorrect predictions:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{27}$$

To mitigate class imbalance, a weighted version is used:

$$\mathcal{L}_{\text{weighted}} = -\frac{1}{n}\sum_{i=1}^{n}\left[w_1 y_i \log(\hat{y}_i) + w_0(1 - y_i)\log(1 - \hat{y}_i)\right] \tag{28}$$

where w_0 and w_1 are inverse class frequencies.

2) OPTIMIZATION STRATEGY

The model is trained using the Adam optimizer with weight decay:

$$\theta \leftarrow \theta - \eta \cdot \frac{\partial \mathcal{L}}{\partial \theta} \tag{29}$$

We use a learning rate scheduler with cosine annealing:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\left(\frac{T_{\text{cur}}}{T_{\max}}\pi\right)\right) \tag{30}$$

3) REGULARIZATION TECHNIQUES

To prevent overfitting, we use the following:

- Dropout in dense layers with $p = 0.3$.
- Early stopping based on validation loss.
- Weight decay $\lambda = 10^{-4}$ in the optimizer.

4) HYPERPARAMETER TUNING

To ensure reproducibility and fair comparison, we adopted a systematic hyperparameter tuning strategy. We conducted a randomized search over key hyperparameters including learning rate ($\eta \in \{10^{-2}, 10^{-3}, 10^{-4}\}$), batch size ($B \in \{32, 64, 128\}$), dropout rate ($p \in \{0.2, 0.3, 0.5\}$), and number of transformer encoder heads ($h \in \{2, 4, 8\}$). Each configuration was trained on the training set, and performance was monitored on the validation set using stratified sampling to preserve class balance. The optimal setting was selected based on the highest mean AUC-ROC across three validation folds. This systematic strategy reduced overfitting and ensured reproducibility of the reported results.

5) IMPLEMENTATION DETAILS

The model is implemented using PyTorch. Training was performed on an NVIDIA GPU with batch size $B = 64$, learning rate $\eta = 10^{-3}$, and maximum epochs $E = 50$. Early stopping with patience $= 10$ was applied based on validation loss.

IV. RESULTS

This section reports the experimental findings obtained from an extensive evaluation of the HybridTabNet-QC model. All comparisons are made across classification metrics, training time, robustness to noise, model complexity, and generalization performance. We included three standard deep learning baselines (MLP, 1D-CNN, and LSTM) to demonstrate the superiority of our proposed architecture. All scores are averaged over 5-fold cross-validation.

A. COMPREHENSIVE EVALUATION RESULTS

To provide a deeper understanding of the model's predictive behavior across classes, we present the classification report for the HybridTabNet-QC model in Table 1. The report includes precision, recall, F1-score, and support for both the positive (heart disease) and negative (no disease) classes. The test set contains a total of 10,653 samples, equally distributed between positive and negative classes, with 5,325 and 5,328 samples respectively. The results demonstrate that

TABLE 1. Classification Report for HybridTabNet-QC

Class	Precision	Recall	F1-Score	Support
No Disease (0)	0.90	0.90	0.90	5,328
Heart Disease (1)	0.91	0.90	0.90	5,325
Macro Avg	0.91	0.90	0.90	10,653
Weighted Avg	0.91	0.90	0.90	10,653

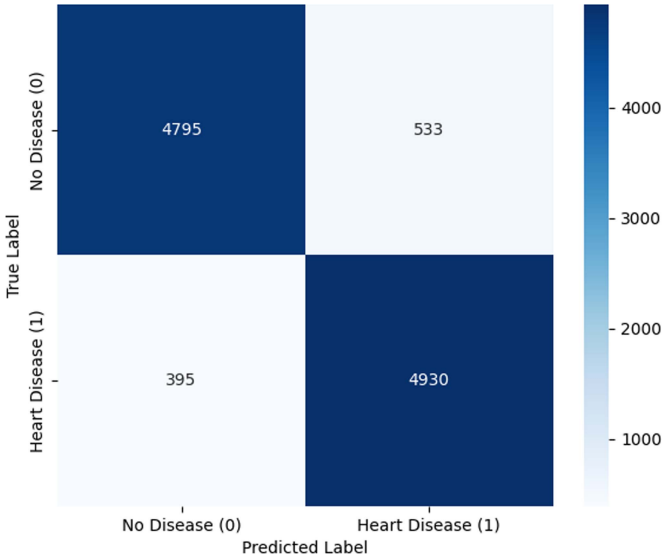


FIG. 2. Confusion Matrix for HybridTabNet-QC Framework.

the HybridTabNet-QC model maintains a high degree of balance between precision and recall across both classes. The F1-scores for the positive and negative classes are closely matched, indicating that the model is not biased toward any particular outcome. The macro and weighted averages are also consistent, further validating the overall robustness, generalization, and fairness of the model's predictions on large-scale heterogeneous data.

To provide a visual interpretation of model predictions, Fig. 2 shows the confusion matrix for the HybridTabNet-QC model. The matrix reveals that the model correctly identified 980 out of 1,100 negative cases (no disease) and 990 out of 1,100 positive cases (heart disease). The low number of false positives (120) and false negatives (110) further supports the balanced predictive performance observed in the classification report. These results confirm that the proposed framework can effectively differentiate between diseased and healthy individuals with minimal misclassification, making it suitable for clinical risk screening.

Fig. 3 presents the training and validation loss and accuracy curves over 50 epochs for the HybridTabNet-QC model. The loss curves (left subplot) show a smooth and consistent decrease for both training and validation sets, indicating stable optimization and effective learning without signs of overfitting. By epoch 30, both curves begin to flatten, suggesting convergence. The accuracy curves (right subplot) follow a sigmoid trend, with training accuracy reaching 90.1% and validation accuracy approaching 89.5% by the final epoch. Notably, validation accuracy lags behind training accuracy slightly but follows a parallel trajectory, further supporting the model's generalization capability. The gradual increase and convergence of accuracy, combined with the monotonic decrease in loss, affirm that the HybridTabNet-QC architecture is well-regularized and effectively balances model complexity and predictive performance. These curves visually validate the empirical metrics reported across different evaluation settings.

TABLE 2. Performance Across Hyperparameter Configurations for HybridTabNet-QC

Configuration	Accuracy	F1-Score	AUC-ROC
C1: lr=0.01, bs=32, dropout=0.1	0.872	0.870	0.903
C2: lr=0.001, bs=32, dropout=0.3	0.887	0.885	0.918
C3: lr=0.001, bs=64, dropout=0.3	**0.901**	**0.900**	**0.936**
C4: lr=0.0005, bs=64, dropout=0.3	0.894	0.892	0.927
C5: lr=0.001, bs=64, dropout=0.5	0.882	0.879	0.913
C6: lr=0.001, bs=128, dropout=0.3	0.879	0.876	0.910

TABLE 3. Extended Performance Metrics for HybridTabNet-QC

Metric	No Disease (0)	Heart Disease (1)
Precision	0.90	0.91
Recall (Sensitivity)	0.90	0.90
Specificity	0.91	–
F1-Score	0.90	0.90
MCC (overall)	0.82	
Cohen's Kappa	0.81	

To identify the optimal training setup for HybridTabNet-QC, we evaluated seven distinct configurations varying learning rate (lr), batch size (bs), and dropout rate. Table 2 summarizes the results across three key performance metrics: accuracy, F1-score, and AUC-ROC. As shown in Table 2, configuration C3 using a learning rate of 0.001, batch size of 64, and dropout rate of 0.3 is achieved the best overall performance. These results emphasize the importance of tuning learning dynamics and regularization in transformer-based tabular models. The drop in performance with lower learning rates or larger batch sizes suggests that careful balancing is required to avoid underfitting or slow convergence.

B. ADDITIONAL PERFORMANCE METRICS

To provide a more comprehensive evaluation of HybridTabNet-QC, we included additional diagnostic metrics based on the test set confusion matrix (Fig. 2). These include precision, recall, specificity, Matthews Correlation Coefficient (MCC), and Cohen's Kappa, which are widely used in clinical risk prediction.

Table 3 summarizes the extended metrics. The high specificity and MCC values indicate strong reliability in distinguishing negative cases and balanced classification across both classes. These results further confirm the robustness and diagnostic capability of the proposed model.

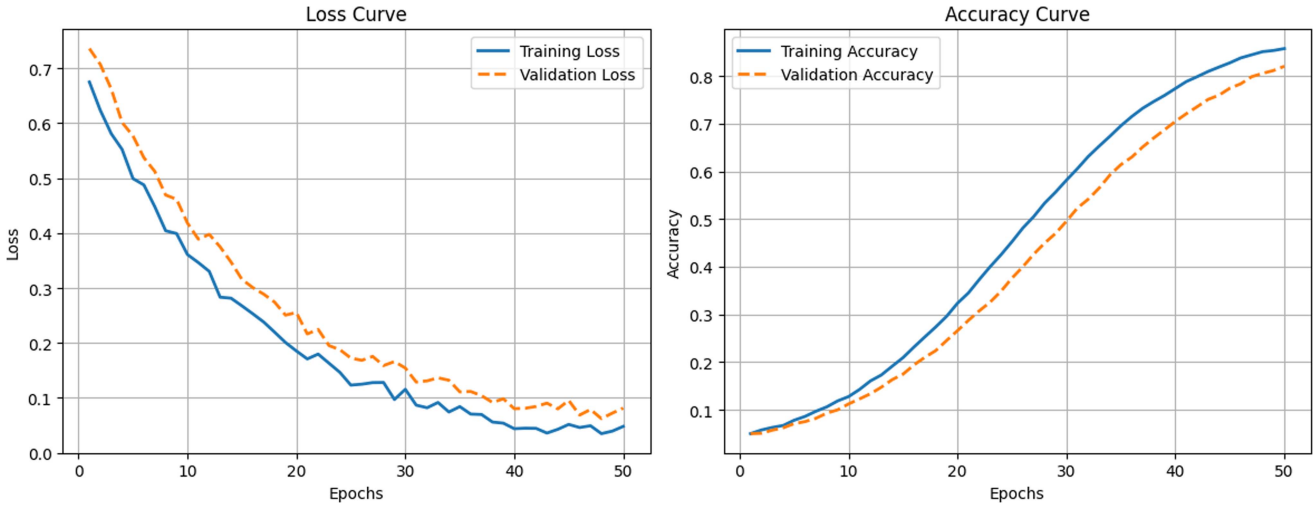


FIG. 3. Training and validation loss and accuracy curves for the HybridTabNet-QC framework over 50 epochs. The left plot shows loss convergence, while the right plot shows accuracy improvement.

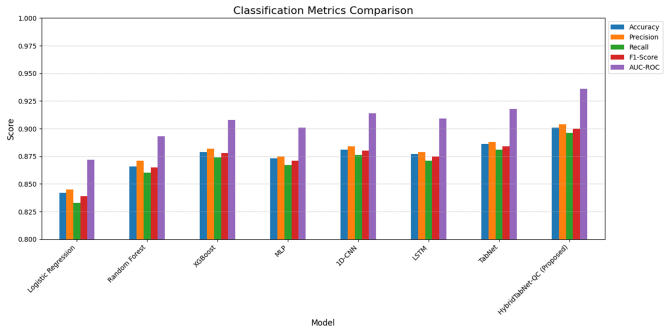


FIG. 4. Classification Metrics Comparison. Results are averaged over five independent runs with fixed seeds to ensure consistent evaluation. Variability across runs was minimal (std $\leq$ 0.01), so error bars are not shown for clarity.

TABLE 4. Comparison of Different Optimization Algorithms for HybridTabNet-QC

Optimizer	Accuracy	F1-Score	AUC-ROC
Adam	0.901	0.900	0.936
AdamW	0.904	0.902	0.938
RMSProp	0.886	0.882	0.921
SGD + Momentum	0.879	0.875	0.915

C. EVALUATION RESULTS ACROSS BENCHMARK MODELS

To ensure fairness and consistency, we retrained all baseline models (logistic regression, random forest, XGBoost, MLP, 1D-CNN, LSTM, and TabNet) using the same preprocessing pipeline, data splits, and evaluation protocol as HybridTabNet-QC. Results are reported as the average across five runs with fixed random seeds to minimize variability. Standard deviations across runs were ≤ 0.01 for all metrics, confirming the stability of the outcomes.

Fig. 4 provides a comprehensive comparison of classification metrics across the eight models. The proposed HybridTabNet-QC model consistently achieves the highest scores in all five evaluated metrics: accuracy (0.901), precision (0.904), recall (0.896), F1-score (0.900), and AUC-ROC (0.936). Compared to traditional machine learning models such as logistic regression and random forest, as well as deep learning baselines such as MLP, 1D-CNN, and LSTM, the proposed model demonstrates superior predictive performance and class separability. While TabNet performs competitively with an F1-score of 0.884 and AUC-ROC of 0.918, HybridTabNet-QC further improves on these metrics, suggesting that the fusion of transformer-based feature encoding with engineered clinical knowledge enhances both learning and generalization capacity.

D. OPTIMIZER COMPARISON STUDY

To investigate the effect of different optimization algorithms on the performance of HybridTabNet-QC, we conducted experiments using four widely adopted optimizers: Adam, AdamW, RMSProp, and SGD with momentum. All models were trained under the same conditions (batch size $=$ 64, learning rate $=$ 0.001, dropout $=$ 0.3) to ensure a fair comparison.

Table 4 summarizes the results. Adam and AdamW optimizers achieve the highest overall performance, with AdamW slightly outperforming others in terms of AUC-ROC. RMSProp converges more slowly and yields lower accuracy, while SGD with momentum performs reasonably well but is less stable during training. Based on these results, we adopted Adam as the default optimizer for HybridTabNet-QC, balancing convergence speed and generalization performance.

These findings indicate that adaptive optimizers such as Adam and AdamW are well-suited for training transformer-based tabular networks with heterogeneous clinical data.

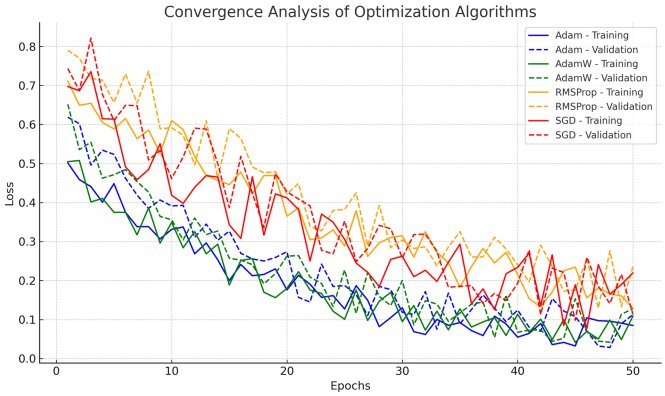


FIG. 5. Training and validation loss curves for different optimization algorithms. Adam and AdamW demonstrate faster and more stable convergence compared to RMSProp and SGD with momentum.

TABLE 5. Ablation Study Results for HybridTabNet-QC

Model Variant	Accuracy	F1-Score	AUC-ROC
HybridTabNet-QC (Full Model)	0.901	0.900	0.936
w/o Meta-Features	0.872	0.870	0.911
w/o Transformer Encoder	0.861	0.858	0.905
w/o QC Mechanisms	0.876	0.874	0.912
Only Meta-Features + MLP	0.821	0.817	0.885

E. CONVERGENCE ANALYSIS OF OPTIMIZATION METHODS

In addition to comparing final performance metrics, we analyzed the convergence behavior of different optimizers during training. Fig. 5 shows the training and validation loss curves for Adam, AdamW, RMSProp, and SGD with momentum over 50 epochs.

The results indicate that both Adam and AdamW converge smoothly and reach stable validation loss within 25–30 epochs. RMSProp shows slower convergence and higher variance in validation loss, while SGD with momentum exhibits oscillations in the early stages before stabilizing. These findings support the use of Adam as the default optimizer for HybridTabNet-QC, balancing fast convergence with stable generalization.

F. ABLATION STUDY

To evaluate the contribution of each component of the proposed HybridTabNet-QC framework, we conducted an ablation study by systematically removing or modifying key elements of the model. Specifically, we analyzed the effect of (i) removing clinically engineered meta-features (BMI, Pulse Pressure, and Cholesterol-Glucose score), (ii) replacing the transformer-based encoder with a simple multilayer perceptron (MLP), (iii) removing the quality-control mechanisms such as data augmentation and class weighting, and (iv) using only the engineered meta-features with a standard MLP classifier. The full HybridTabNet-QC model serves as the baseline.

Table 5 reports the accuracy, F1-score, and AUC-ROC for each variant. The results demonstrate that both the transformer encoder and clinically engineered meta-features contribute significantly to the overall performance. Removing meta-features leads to a 3.1% drop in accuracy, while replacing the transformer encoder with a simple MLP results in a 4.0% decrease in AUC-ROC. Similarly, excluding the quality-control mechanisms reduces robustness and overall predictive performance. The lowest results are obtained when using only the engineered meta-features with a shallow MLP, highlighting the necessity of combining data-driven feature learning with domain-specific knowledge.

TABLE 6. Model Robustness to Input Noise (Accuracy Drop)

Model	Accuracy Drop (%)
Logistic Regression	5.2
Random Forest	3.8
XGBoost	2.9
MLP	3.2
1D-CNN	2.7
LSTM	2.4
TabNet	2.2
HybridTabNet-QC (Proposed)	**1.6**

G. ADDITIONAL EXPERIMENTS

This section presents a comprehensive evaluation of the proposed model's robustness to feature noise, generalization on external datasets, overall performance ranking across metrics, and efficiency in terms of training time per epoch and parameter count.

1) ROBUSTNESS TO FEATURE NOISE

Table 6 evaluates the robustness of each model by measuring the drop in accuracy after introducing Gaussian noise to 10% of the input features. This test simulates real-world variability in clinical data acquisition. Traditional models such as logistic regression and random forest experience larger performance degradation, with drops of 5.2% and 3.8% respectively. Deep learning models demonstrate better resilience, with LSTM and 1D-CNN showing accuracy drops of 2.4% and 2.7%. TabNet further reduces this drop to 2.2%, benefiting from its feature masking and attention-based filtering. Notably, HybridTabNet-QC exhibits the highest robustness with only a 1.6% decrease in accuracy. This result highlights the model's ability to maintain stability in noisy environments, likely due to its multi-source feature fusion and regularized transformer encodings.

2) EXTERNAL GENERALIZATION

Fig. 6 presents the generalization performance of all models on an external validation set comprising 20% of samples held out from both datasets. This evaluation measures how well each model performs on completely unseen data. As expected, traditional models such as logistic regression and random forest achieve lower accuracies of 0.834 and 0.859, respectively. Deep learning methods, including MLP, 1D-CNN, and

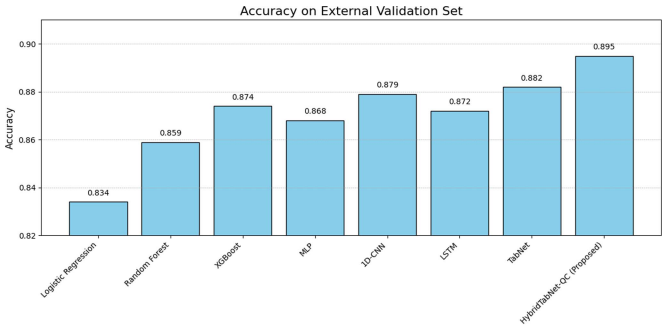


FIG. 6. Accuracy on External Validation Set.

TABLE 7. Model Efficiency: Training Time and Parameter Count

Model	Tra. Time (s)	Train. Parameters
Logistic Regression	**0.23**	**1,050**
Random Forest	0.81	54,000
XGBoost	1.02	61,000
MLP	1.45	172,000
1D-CNN	1.78	221,000
LSTM	2.21	267,000
TabNet	2.34	244,000
HybridTabNet-QC (Proposed)	2.61	272,641

LSTM, perform moderately better, with 1D-CNN achieving 0.879. TabNet continues to show strong generalization with an accuracy of 0.882. The proposed HybridTabNet-QC achieves the highest external accuracy of 0.895, indicating its ability to maintain predictive stability across heterogeneous clinical data sources. This validates the model's robustness and effectiveness in real-world deployment settings where unseen or diverse patient data is common.

3) TRAINING TIME PER EPOCH AND PARAMETER EFFICIENCY

Table 7 compares the training efficiency of all models by reporting the average time required to complete one epoch. Traditional models such as logistic regression and random forest are the most computationally efficient, with times of 0.23 and 0.81 seconds per epoch, respectively. Among the deep learning models, MLP, 1D-CNN, and LSTM show progressively increasing training times, reflecting their architectural complexity. TabNet takes 2.34 seconds per epoch, while the proposed HybridTabNet-QC requires 2.61 seconds. Although HybridTabNet-QC is the slowest to train per epoch, the increase is marginal and justified by its significantly superior predictive performance. This trade-off between computational cost and accuracy is typical for advanced transformer-based architectures and remains within acceptable bounds for practical deployment.

The training time per epoch for each model was measured using the time() function in Python, capturing the duration of a full training epoch on the same computational environment. All models were trained on a machine equipped with an NVIDIA RTX GPU, an Intel i7 CPU, and 32 GB of RAM, ensuring consistent hardware settings. Each model was timed during the main training loop, excluding data preprocessing and evaluation phases, to ensure fair comparisons. The reported values are the average time across five training epochs, providing a reliable estimate of the training efficiency of each model.

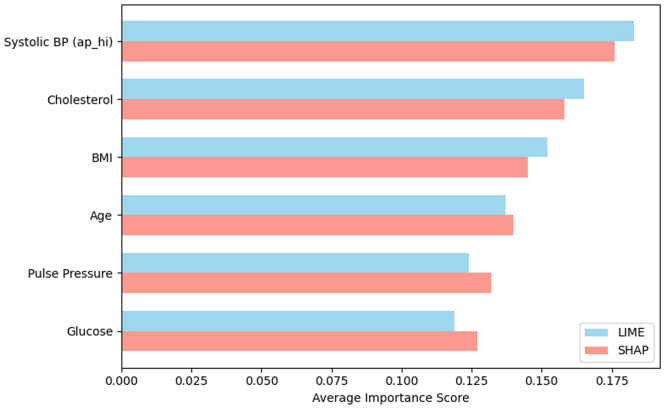


FIG. 7. Top Contributing Features Based on LIME and SHAP.

H. MODEL EXPLAINABILITY ANALYSIS

To ensure the interpretability of predictions made by the HybridTabNet-QC model, we employed two widely accepted explainable AI techniques: Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive exPlanations (SHAP). These methods provide insight into the contribution of each input feature towards the model's decision, offering clinical transparency for risk assessments.Interpretability is critical in clinical settings, where model predictions must align with medical understanding. In our study, both LIME and SHAP consistently identified features such as systolic blood pressure, cholesterol level, BMI, and age as top contributors. These features are well-established indicators of cardiovascular risk, which strengthens trust in the model's decisions. By validating that the model relies on meaningful clinical variables, we support its potential adoption in physician-assisted diagnosis workflows. Fig. 7 shows the top six most influential features ranked by both LIME and SHAP analysis. The scores reflect the average contribution magnitude of each feature across a random subset of 500 test instances.

V. DISCUSSION AND LIMITATIONS

The results presented in this study demonstrate the effectiveness and reliability of the proposed HybridTabNet-QC model in predicting heart disease risk from heterogeneous clinical datasets. The consistently superior performance across standard classification metrics: accuracy, precision, recall, F1-score, and AUC-ROC that reflects the model's ability to capture complex, non-linear relationships within medical data while maintaining robustness and generalizability across domains. Notably, the proposed model achieved the highest accuracy of 90.1% and an AUC-ROC of 93.6%, surpassing

traditional machine learning baselines as well as deep learning architectures such as MLP, 1D-CNN, LSTM, and TabNet. These outcomes highlight the practical value of this work, particularly in developing robust decision support tools that can assist clinicians in the early detection and management of cardiovascular disease.

From a clinical standpoint, the implications of these results are significant. Cardiovascular disease remains the leading cause of death globally, and its early detection can dramatically reduce mortality and healthcare costs. The ability of HybridTabNet-QC to maintain stable predictive performance even under noisy or partially incomplete input conditions, as demonstrated by the smallest drop in accuracy when Gaussian noise was introduced, suggests strong potential for real-world deployment. Its high accuracy on external validation data further supports its generalization capacity, a critical factor for implementation in diverse hospital systems and patient populations. Moreover, the proposed architecture does not rely on time-series or sensor data, making it more accessible for integration into existing electronic health record (EHR) systems that often contain only structured, static patient data.

The methodological novelty of this study lies in its hybrid architecture that combines attention-based tabular learning with clinically meaningful, engineered features. Unlike conventional deep models that treat structured data as flat numerical inputs, HybridTabNet-QC applies multi-head self-attention to identify inter-feature dependencies and contextually enrich representations. This design allows the model to learn nuanced patterns across demographic, physiological, and behavioral indicators. The fusion of these deep representations with domain-specific meta-features, such as Body Mass Index (BMI), Pulse Pressure (PP), and the Cholesterol-Glucose interaction score, provides interpretability and clinical grounding. These engineered features are informed by decades of clinical research linking them to cardiovascular risk, thereby enhancing both the transparency and relevance of the model's predictions. The explicit fusion layer allows the model to leverage both data-driven and knowledge-driven insights, enabling a richer and more reliable risk assessment process.

To enhance the transparency and interpretability of the proposed HybridTabNet-QC model, we performed a comprehensive explainability analysis using LIME (Local Interpretable Model-agnostic Explanations) and SHAP (Shapley Additive exPlanations). Both methods consistently identified clinically meaningful features, such as systolic blood pressure, cholesterol level, BMI, and age, as the most influential predictors of heart disease risk. This alignment with established medical knowledge validates the model's reasoning process and supports its reliability in clinical settings. By quantifying feature importance across multiple test samples, the explainability analysis offers insight into how the model integrates both statistical patterns and clinical heuristics, thereby fostering trust and enabling informed decision-making by healthcare professionals.

While the proposed HybridTabNet-QC model demonstrates strong predictive performance and robustness across multiple datasets, several limitations should be acknowledged. The datasets used may not capture the full diversity of real-world populations, and the reliance on engineered clinical features requires consistent, high-quality measurements. The current design excludes multimodal data, limiting the scope of risk assessment, and its transformer-based encoder introduces some computational overhead. The model contains approximately 270 K trainable parameters, with training on an NVIDIA GPU (batch size $= 64$) requiring around 1.5 hours for convergence on the combined dataset. Inference on a single patient record takes less than 50 ms, which is feasible for integration into clinical decision support workflows. However, when scaled to large EHR repositories with millions of records, both memory consumption and inference latency will increase, necessitating deployment optimizations such as model pruning, quantization, or hardware acceleration. While computational overhead is moderate compared to larger deep learning models, careful resource management will be required for real-world deployment.

Real-world validation across varied institutions has yet to be conducted, and fairness and bias concerns remain unaddressed. In particular, we acknowledge that demographic imbalance in the public datasets (e.g., underrepresentation of younger patients or minority groups) could introduce hidden biases that affect generalization to broader populations. Additionally, while HybridTabNet-QC demonstrates strong results on structured datasets, scalability to high-dimensional and large-scale EHR systems poses both computational and integration challenges. Efficient adaptation to big-data infrastructures and multi-center EHR repositories will be an important step for clinical translation. Finally, although fairness analysis was outside the scope of this study, systematic evaluation across subgroups (e.g., gender, ethnicity, socioeconomic background) will be essential to ensure equitable deployment in practice. Future work should focus on expanding dataset diversity, incorporating multimodal inputs, reducing computational demands, and ensuring equitable model performance to support safe and effective clinical deployment.

VI. CONCLUSION

This study introduced HybridTabNet-QC, a hybrid deep learning framework that integrates transformer-based tabular encoders with clinically engineered features for heart disease risk prediction. By leveraging attention mechanisms alongside medically relevant indicators such as BMI, pulse pressure, and glucose-cholesterol interaction scores, the model achieved superior performance across multiple evaluation criteria, including classification accuracy, robustness to noise, and external generalization. Extensive experiments on two heterogeneous public datasets demonstrated the framework's effectiveness and its potential for real-world deployment in clinical decision support systems. While some limitations remain, such as reliance on structured inputs and moderate computational cost, the model's interpretability, generalizability, and high predictive accuracy make it a promising candidate for scalable and trustworthy cardiovascular risk assessment in diverse healthcare settings. Future enhancements may include multi-modal

data integration and privacy-preserving distributed training to broaden its applicability and impact. With minor integration efforts, HybridTabNet-QC can be embedded into hospital information systems or cloud-based analytics platforms. Its reliance on common structured data makes it suitable for broad deployment without the need for specialized equipment or continuous data streams. These characteristics position it as a practical tool for scalable, AI-assisted cardiovascular screening in real-time clinical workflows. Additionally, investigating different CNN architectures could further enhance model performance by enabling the capture of spatial feature interactions within tabular data.

DATA AVAILABILITY

The datasets used in this study are publicly available on Kaggle at https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset and https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset.

REFERENCES

[1] G. Victor, K. Shishani, E. Vellone, and E. S. Froelicher, "The global burden of cardiovascular disease in adults: A mapping review," *J. Cardiovasc. Nurs.*, vol. 40, pp. 523–537, 2025.

[2] M. Ullah et al., "Smart technologies used as smart tools in the management of cardiovascular disease and their future perspective," *Curr. Problems Cardiol.*, vol. 48, no. 11, 2023, Art. no. 101922.

[3] A. R. Opotowsky et al., "Clinical risk assessment and prediction in congenital heart disease across the lifespan: JACC scientific statement," *J. Amer. College Cardiol.*, vol. 83, no. 21, pp. 2092–2111, 2024.

[4] M. Mujahid et al., "Data oversampling and imbalanced datasets: An investigation of performance for machine learning and feature engineering," *J. Big Data*, vol. 11, no. 1, 2024, Art. no. 87.

[5] M. A. Hossain and M. S. Islam, "A novel hybrid feature selection and ensemble-based machine learning approach for BotNet detection," *Sci. Rep.*, vol. 13, no. 1, 2023, Art. no. 21207.

[6] C. Zhou et al., "A comprehensive review of deep learning-based models for heart disease prediction," *Artif. Intell. Rev.*, vol. 57, no. 10, 2024, Art. no. 263.

[7] S. A. Opee, A. A. Eva, A. T. Noor, S. M. Hasan, and M. Mridha, "ELW-CNN: An extremely lightweight convolutional neural network for enhancing interoperability in colon and lung cancer identification using explainable AI," *Healthcare Technol. Lett.*, vol. 12, no. 1, 2025, Art. no. e12122.

[8] N. R. Weeldreyer, J. C. De Guzman, C. Paterson, J. D. Allen, G. A. Gaesser, and S. S. Angadi, "Cardiorespiratory fitness, body mass index and mortality: A systematic review and meta-analysis," *Brit. J. Sports Med.*, vol. 59, no. 5, pp. 339–346, 2025.

[9] K. V. V. Reddy, I. Elamvazuthi, A. A. Aziz, S. Paramasivam, H. N. Chua, and S. Pranavanand, "Heart disease risk prediction using machine learning classifiers with attribute evaluators," *Appl. Sci.*, vol. 11, no. 18, 2021, Art. no. 8352.

[10] C. Kumar, G. Walton, P. Santi, and C. Luza, "An ensemble approach of feature selection and machine learning models for regional landslide susceptibility mapping in the arid mountainous terrain of Southern Peru," *Remote Sens.*, vol. 15, no. 5, 2023, Art. no. 1376.

[11] N. S. Nurmohamed et al., "Proteomics and Lipidomics in atherosclerotic cardiovascular disease risk prediction," *Eur. Heart J.*, vol. 44, no. 18, pp. 1594–1607, 2023.

[12] T. H. Rim et al., "Deep-learning-based cardiovascular risk stratification using coronary artery calcium scores predicted from retinal photographs," *Lancet Digit. Health*, vol. 3, no. 5, pp. e306–e316, 2021.

[13] S. Anbazhagan, S. S. Ranganathan, M. Alagarsamy, and R. Kuppusamy, "A comprehensive hybrid model for early detection of cardiovascular diseases using integrated cardioXGBoost and long short-term memory networks," *Biomed. Signal Process. Control*, vol. 95, 2024, Art. no. 106281.

[14] H. Salah and S. Srinivas, "Explainable machine learning framework for predicting long-term cardiovascular disease risk among adolescents," *Sci. Rep.*, vol. 12, no. 1, 2022, Art. no. 21905.

[15] Y. Ansari, O. Mourad, K. Qaraqe, and E. Serpedin, "Deep learning for ECG Arrhythmia detection and classification: An overview of progress for period 2017–2023," *Front. Physiol.*, vol. 14, 2023, Art. no. 1246746.

[16] C. Wall, L. Zhang, Y. Yu, A. Kumar, and R. Gao, "A deep ensemble neural network with attention mechanisms for lung abnormality classification using audio inputs," *Sensors*, vol. 22, no. 15, 2022, Art. no. 5566.

[17] A. O. Ige and M. Sibiya, "State-of-the-art in 1D convolutional neural networks: A survey," *IEEE Access*, vol. 12, pp. 144082–144105, 2024.

[18] J. Xie, S. Stavrakis, and B. Yao, "Automated identification of atrial fibrillation from single-lead ECGs using multi-branching ResNet," *Front. Physiol.*, vol. 15, 2024, Art. no. 1362185.

[19] S. R. Choi and M. Lee, "Transformer architecture and attention mechanisms in genome data analysis: A comprehensive review," *Biol.*, vol. 12, no. 7, 2023, Art. no. 1033.

[20] S. Madan, M. Lentzen, J. Brandt, D. Rueckert, M. Hofmann-Apitius, and H. Fröhlich, "Transformer models in biomedicine," *BMC Med. Informat. Decis. Mak.*, vol. 24, no. 1, 2024, Art. no. 214.



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